



Urban Informatics

Assignment - Week 12

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total marks: (10 X 2) = 20 marks

MCQ/MSQ Question

QUESTION 1: [2 marks]

What is the primary function of a servo motor when used with an Arduino?

- a. Precise control of angular position based on input signals
- b. Continuous high-speed rotation without control
- c. Operating only at fixed speeds
- d. Producing linear motion without any rotation

Correct Answer: a. Precise control of angular position based on input signals

Detailed explanation: Refer to lecture 56, slide 3.

QUESTION 2: [2 marks]

How does a Passive Infrared (PIR) sensor such as the HC-SR501 detect motion?

- a. Measuring changes in air pressure
- b. Detecting sound frequency variations in the environment
- c. Sensing changes in infrared radiation from warm objects
- d. Emitting ultrasonic waves and measuring their reflection

Correct Answer: c. Sensing changes in infrared radiation from warm objects

Detailed explanation: Refer to lecture 56, slide 6.

QUESTION 3: [2 marks]

In GIS, a query is defined as:

- a. A process of data deletion
- b. A method to draw maps manually
- c. A technique used to design computer hardware components
- d. A request to select features or records from a database

Correct Answer: d. A request to select features or records from a database

Detailed explanation: Refer to lecture 57, slide 3.



QUESTION 4: [2 marks]

What is the primary requirement, when joining a CSV file to a shapefile in GIS?

- a. Both files must have identical geometry types
- b. The CSV file must contain spatial coordinates
- c. Both datasets must contain a common matching field
- d. The shapefile must be converted into raster format

Correct Answer: c. Both datasets must contain a common matching field

Detailed explanation: Refer to lecture 57, slide 19.

QUESTION 5: [2 marks]

When is the “Relate” function most useful in GIS?

- a. When both tables have identical records
- b. When a one-to-many relationship exists between tables
- c. When spatial data needs to be visualized
- d. When converting coordinate systems

Correct Answer: b. When a one-to-many relationship exists between tables

Detailed explanation: Refer to lecture 57, slide 22.

QUESTION 6: [2 marks]

What is the purpose of classification in forest-based models?

- a. Predicting continuous values
- b. Grouping data without labels
- c. Predicting categorical variables
- d. Reducing dimensionality

Correct Answer: c. Predicting categorical variables

Detailed Solution: Refer to lecture 58, slide 13.

QUESTION 7: [2 marks]

Which tool is used to evaluate classification performance?

- a. Confusion Matrix
- b. Buffer Tool
- c. Scatter Plot
- d. Histogram

Correct Answer: a. Confusion Matrix

Detailed Solution: Refer to lecture 58, slide 13.



QUESTION 8: [2 marks]

Which of the following values indicate the presence of a hotspot in Hot Spot Analysis (Getis-Ord G_i^*)?

- a. Positive statistically significant z-scores
- b. Zero z-scores
- c. Random distribution values
- d. Negative z-scores

Correct Answer: a. Positive statistically significant z-scores

Detailed Solution: Refer to lecture 47, slide 10.

QUESTION 9: [2 marks]

Why is the “Create Fishnet” tool used before Hot Spot Analysis?

- a. To convert raster data into vector format
- b. To create a polygon grid for aggregating point data
- c. To remove spatial outliers
- d. To calculate distance automatically

Correct Answer: b. To create a polygon grid for aggregating point data

Detailed Solution: Refer to lecture 59, slide 4.

QUESTION 10: [2 marks]

What is the main use of the ArcGIS API for Python?

- a. Hardware design
- b. Desktop-only spatial analysis
- c. Image editing
- d. Web GIS, cloud-based mapping and data access

Correct Answer: d. Web GIS, cloud-based mapping and data access

Detailed Solution: Refer to lecture 60, slide 8.

*****END*****